# Pratik Nanaso Raut

Pune, Maharashtra | Open to relocation -- Jaipur | +91 8600475940 | pratikraut.codex@gmail.com |
https://linkedin.com/in/pratik-raut-85248b319 | https://github.com/pratik-codeX

## PROFESSIONAL SUMMARY

MCA candidate with strong foundation in Java, Spring Boot, and REST APIs — adept at developing robust backend solutions.

Hands-on experience architecting multi-tenant systems using Spring Security and JWT in a SaaS environment with focus on Database Optimization and API Security.

## TECHNICAL SKILLS

| | |
|---|---|
| **Languages** | Java, SQL, JavaScript, C, C++, Python (basics) |
| **Backend** | Spring Boot, REST APIs, Hibernate, Spring Security, JWT, JDBC |
| **Databases** | MySQL, MongoDB |
| **Tools** | Git, GitHub, Postman, Maven, IntelliJ IDEA |
| **Concepts** | Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Problem Solving, SDLC, REST API Design, Database Optimization, API Security, Multi-Tenant Architecture, JWT Authentication, RBAC, Design Patterns, Time Complexity Analysis, Microservice Fundamentals, Multithreading |
| **Exposure** | AWS Fundamentals, AI/ML Fundamentals, Big Data Concepts |

## PROFESSIONAL EXPERIENCE

**Software Developer Intern** *Feb 2026 -- Ongoing*
*SetTribe IT Solutions | Pune*

- Developed and enhanced ERP backend modules using Java, Spring Boot, Hibernate, and REST APIs for employee and operational workflows.
- Integrated PostgreSQL database operations, validation logic, and API testing workflows using Postman.
- Collaborated in Agile sprint cycles, performing Git-based code reviews, bug fixes, feature implementation, and deployment support.

## ACADEMIC PROJECTS

**PayStride** -- Multi-Tenant SaaS Payroll Management System *Java, Spring Boot, MySQL, Hibernate, REST API, JWT*

- Architected multi-tenant RBAC system using Spring Security and JWT, enforcing complete data isolation across all API endpoints.
- Optimized MySQL query execution through strategic indexing and schema normalization, reducing average response latency by 40%.
- Engineered BigDecimal payroll computation engine handling statutory deductions (PF 12%, ESI 0.75%) and 1.5x overtime multipliers with zero precision loss.

**EduTrack** -- Student Management Portal *Java, Spring Boot, MongoDB, REST API*

- Developed RESTful CRUD APIs over Spring Boot handling 500+ student records with optimized MongoDB aggregation queries.
- Designed modular service architecture enabling independent scaling of student, batch, and reporting modules.
- Implemented dual-layer validation (client-side and server-side) reducing data entry errors by 25%.

**Customised Virtual File System (CVFS)** -- In-Memory Linux-like File Manager *C, C++, Data Structures, OS Concepts*

- Simulated Linux file system operations (open, read, write, lseek) in C++ using inode-based metadata management and a custom shell interpreter.
- Achieved O(1) file lookup by designing hash table-based inode indexing with chaining for collision resolution.
- Implemented in-memory file allocation table using C++ classes managing file permissions, ownership, and hierarchical directory traversal.

## TECHNICAL STRENGTHS

- Proficient in backend application development, REST API design, and database-driven systems.
- Hands-on experience building concurrent, multi-tenant, and performance-optimized Java applications.
- Experience in Agile development workflows, Git-based version control, and collaborative engineering.
- Strong debugging, query optimization, and software architecture capabilities.

## EDUCATION

**Master of Computer Applications (MCA)** *2026 | Pursuing*
Dnyansagar Institute of Management and Research, Pune (SPPU) 8.25 CGPA | Sem VII: 7.24 SGPA

**Bachelor of Computer Applications (BCA)** *2024*
C.T. Bora College, Shirur (SPPU) 73.34%

**Higher Secondary Certificate (HSC)** *2021*
Maharashtra State Board 75%